

7 Transport in plants

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Xylem dead lignified vessels carry water upward under transpiration pull; cohesion and adhesion maintain water columns. Phloem living sieve tubes translocate sucrose from sources to sinks, driven by loading, osmotic water movement and pressure flow. Water potential differences at root, leaf and atmosphere sustain net transport.

Worked example: follow the reasoning

A leafy shoot loses 0.24 g water in 4 min. Assuming 1 g mL⁻¹, transpiration rate is 0.060 mL min⁻¹.

Quick check · multiple choice

A Which tissue translocates sucrose from source to sink?

[1]

- A Xylem vessel
- B Epidermis
- C Phloem sieve tubes
- D Palisade cells

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Describe how increasing wind speed may change transpiration rate and why the effect can plateau.

[3]

Answers and explanation

Quick check: C. Phloem sieve tubes

Question 1 [3 marks]

Wind removes humid boundary layer [1], increasing water-vapour gradient and evaporation [1]; rate may become limited by stomatal aperture/water supply or closure in water stress [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Biology 9700, 2025–2027. <https://www.cambridgeinternational.org/Images/664560-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.